Reward learning, perseveration and reflection in small language models: a cognitive-modelling comparison with humans

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Abstract.
Large language models (LLMs) are increasingly studied as if they were participants in psychology experiments. I applied this approach to three small open models (llama3.2:3b, qwen2.5:3b and qwen2.5:7b) that run on a laptop, using two classic paradigms. In a two-armed bandit, I fitted Q-learning, Q-learning with perseveration, win-stay lose-shift and random models to each participant by maximum likelihood, and compared the models with 44 human participants from Lefebvre et al. (2017) taken from the Psych-101 collection. Humans learned quickly (median 82% correct), and most were best described by Q-learning. Neither 3B model learned. llama3.2 stayed at chance, and qwen2.5:3b was dominated by perseveration: it switched on only 7% of trials, and its stickiness parameter κ was six times the human value. qwen2.5:7b did learn, reaching a median of 90% correct on the human task design (humans: 82%), but it was just as sticky as the 3B model and was best described by Q-learning with perseveration. Scale thus added reward learning without removing the habit. On the Cognitive Reflection Test, llama3.2 outperformed the human average on the three famous items but solved only 18% of six new items with the same structure, a pattern consistent with memorisation. qwen2.5:7b showed the reverse: it solved 97% of the new items but gave the intuitive answer to the famous bat-and-ball problem on every sample. Perseveration had to be modelled explicitly: without it, structured LLM behaviour was misclassified as random. All code, data and analyses are openly available.

Keywords: large language models; reinforcement learning; multi-armed bandit; perseveration; cognitive reflection; computational modelling

1. Introduction

Cognitive psychology has a long tradition of explaining behaviour with formal models that are fitted to individual participants. Binz and Schulz (2023) proposed applying the same tools to large language models (LLMs). The model is treated as a participant, given a standard task, and then analysed with the models used for people. This approach asks *how* a model behaves, not only whether its answers are correct. Since then, LLMs have been tested on batteries of cognitive tasks (Coda-Forno et al., 2024), on reasoning biases such as those in the Cognitive Reflection Test (Hagendorff et al., 2023), and on reinforcement-learning tasks in which their in-context learning can be described by asymmetric belief updating (Schubert et al., 2024).

Most of this work studies large commercial models. Small open models (a few billion parameters) are widely used, cheap to run and fully reproducible, but much less is known about their cognitive profile. They are also a useful test case. If behaviours such as reward learning appear only at large scale, small models should fail in ways that cognitive models can describe.

This study asks two questions:

1. **Learning from reward.** In a two-armed bandit, do small LLMs learn which option pays more? Is their behaviour captured by the reinforcement-learning model that describes humans, and how do their parameters compare?
2. **Reflection versus intuition.** On the Cognitive Reflection Test (CRT; Frederick, 2005), do LLMs make the same intuitive errors as people? Do they solve the famous items only because those items appear in their training data?

I expected the models to learn the bandit while exploring less than humans, and to solve the original CRT items more often than new items with the same structure.

2. Methods

2.1 Participants

LLMs. I used llama3.2:3b, qwen2.5:3b and qwen2.5:7b, all run locally through Ollama. Including two sizes of the same model family allows a first look at the effect of scale. Each model contributed 20 “participants”, which were the same model sampled with different random seeds. Sampling temperature was 1.0 in the bandit task and 0.7 in the CRT. Answers were limited to 5 tokens. Unparseable answers were retried up to 3 times and then replaced by a random choice.

No choice from any model needed this fallback (0 of 6,000 per model). A third model, qwen3:4b, was excluded after piloting. Its standard version always produces reasoning text before answering, and its instruct version exceeded the answer limit on 96 of 240 pilot trials.

Humans. I used 44 participants from experiment 1 of Lefebvre et al. (2017), taken from the Psych-101 collection (Binz et al., 2025).

Reference learner. I simulated 20 Q-learners on the LLM task, to show what a genuine reward learner looks like under the same analysis.

2.2 Two-armed bandit

LLM task. Each participant played 10 games of 30 trials. In each game, two slot machines paid 1 point with hidden probabilities drawn uniformly between 0.1 and 0.9, at least 0.2 apart. Following Binz and Schulz (2023), the prompt described a casino with machines labelled F and J. It then listed the full history of the current game (e.g. "Round 3: you played Machine F and won 1 point.") and asked for a single letter. The assignment of letters to machines was randomised between games, so a preference for one letter could not look like learning.

Human task. In Lefebvre et al. (2017), participants visited four casinos 24 times each, interleaved. Each casino had two machines that paid 0 or 0.5 points. Pooled over participants, the payout rates were 25/25%, 75/25%, 25/75% and 75/75%. Only the two asymmetric casinos have a better option, so only those were analysed, as 2 games of 24 trials (2,112 choices in total). A payout of 0.5 was coded as reward 1.

Matched design. The LLM task is harder than the human one, so all three models also played the human design: 20 participants, 2 games of 24 trials, with machines paying 75% versus 25% (0 of 960 fallbacks per model). Unlike the humans, the LLMs played each game in a single block rather than interleaved.

2.3 Cognitive models

I fitted four models to each participant by maximum likelihood (L-BFGS-B, 10 random restarts) and compared them with the Bayesian information criterion (BIC), following Wilson and Collins (2019).

- **Q-learning.** After choosing option c and receiving reward r, $Q_c \leftarrow Q_c + \alpha(r - Q_c)$, with Q initialised at 0.5. Choices follow a softmax, $P(c) \propto \exp(\beta\, Q_c)$. α is the learning rate and β the inverse temperature.
- **Q-learning with perseveration.** $P(c) \propto \exp(\beta\, Q_c + \kappa \cdot 1[c = \text{previous choice}])$. $\kappa > 0$ is a tendency to repeat the previous choice regardless of reward (Lau & Glimcher, 2005; Katahira, 2015). I added this model after a first analysis showed that qwen2.5 almost never switched, which none of the other models captured.
- **Win-stay lose-shift (WSLS)**, with a noise parameter ε.
- **Random**, which chooses each option with probability 0.5.

α and β were interpreted only for participants best fitted by Q-learning. κ was compared across all participants. Groups were compared with two-sided Mann-Whitney U tests on per-participant medians. I also report model-free measures: accuracy (the proportion of choices of the better option), switch rate, win-stay and lose-shift.

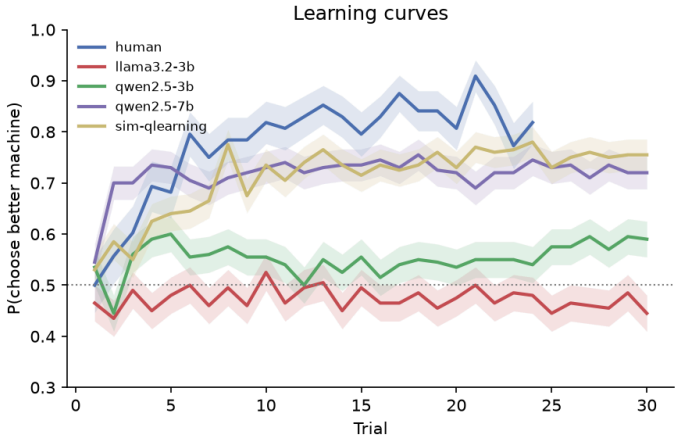
Parameter recovery. I simulated 40 agents with known parameters and refitted them. With the LLM design (300 trials per participant), α and β were well recovered (Spearman $r = 0.96$ and 0.90). With the human design (48 trials), recovery was weak (α: $r = 0.59$; β: $r = 0.69$). Human α and β were therefore not compared directly with the LLMs. κ (drawn from -2 to 4) was well recovered at both designs ($r = 0.98$ at 300 trials, 0.91 at 48 trials), so stickiness can be compared reliably between humans and models.

2.4 Cognitive Reflection Test

The test had nine items: the three originals (Frederick, 2005) and six new items written for this study. Each new item shares the structure of one original but uses new numbers and a new story (two "costs X more" items, two "N machines make N things" items and two "doubling" items). Each item was presented 10 times. Each answer was scored as correct, intuitive (the tempting wrong answer, such as 10 cents in the bat-and-ball problem) or other. The human reference is the published mean of 1.24 of 3 correct (Frederick, 2005).

3. Results

3.1 Bandit: main design



Learning curves on the main LLM design. The proportion of choices of the better option is plotted by trial, alongside human data from Lefebvre et al. (2017) and simulated Q-learners.

Table 1. Bandit behaviour and model fits on the main design (medians per participant). Accuracy is the proportion of choices of the better option.

Group	Accuracy	Switch	Win-stay	Lose-shift	Best model (count)	κ
Humans (n = 44)	0.82	0.21	0.90	0.33	Q 26, Q+P 9, random 8, WSLS 1	0.38
llama3.2:3b (n = 20)	0.50	0.37	0.64	0.33	Q+P 12, random 8	0.55
qwen2.5:3b (n = 20)	0.53	0.07	0.93	0.08	Q+P 20	2.42
qwen2.5:7b (n = 20)	0.72	0.08	0.98	0.15	Q+P 20	2.17
Simulated Q-learner (n = 20)	0.71	0.25	0.89	0.44	Q 20	0.04

Note. Q = Q-learning; Q+P = Q-learning with perseveration.

Humans learned quickly, reaching about 80% correct within six trials (Figure 1). Most humans (26 of 44) were best fitted by plain Q-learning, with median $\alpha = 0.18$ and $\beta = 9.4$ among those participants.

Neither 3B model learned. Both were far less accurate than humans (llama3.2: $U = 826$, $p < 10^{-7}$; qwen2.5:3b: $U = 761$, $p < 10^{-5}$). No LLM participant on this design was best fitted by plain Q-learning.

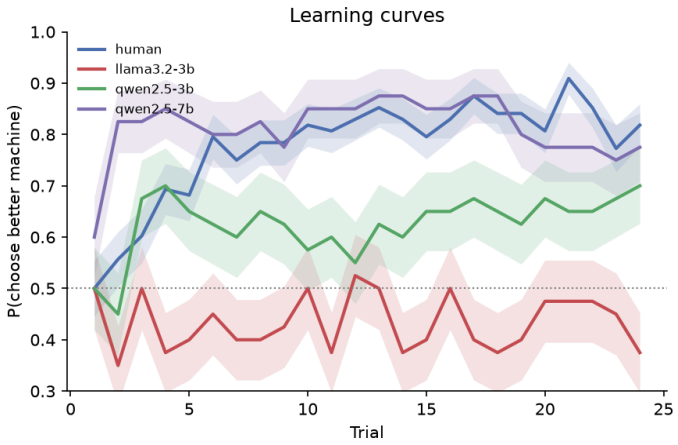
llama3.2 stayed at chance across all 30 trials. For the 12 participants best fitted by the perseveration model, the fitted α and β were both close to zero, so reward had no measurable effect on choice. The remaining 8 participants were best described as random. Compared with humans, llama3.2 switched more often ($U = 203$, $p < .001$) and stayed less often after a win ($U = 762$, $p < 10^{-5}$). Its κ did not differ from the human κ ($U = 421$, $p = .79$).

qwen2.5:3b was extremely sticky. It switched on 7% of trials against 21% for humans ($U = 691$, $p < .001$), and it almost never shifted after a loss (8% vs 33%; $U = 811$, $p < 10^{-7}$). Its κ was about six times the human value (2.42 vs 0.38; $U = 132$, $p < 10^{-5}$), and the perseveration model won for all 20 participants. Its learning curve rose slightly in the first trials and then stayed flat at about 55%.

qwen2.5:7b learned. Its median accuracy (0.72) matched that of the simulated Q-learners, although it remained below humans ($U = 282$, $p = .022$). It was nevertheless as sticky as its 3B counterpart: it switched on 8% of trials (vs 21% for humans; $U = 209$, $p < .001$), shifted after only 15% of losses (vs 33%; $U = 124$, $p < 10^{-5}$), and its κ (2.17) was far above the human value ($U = 134$, $p < 10^{-5}$). The perseveration model won for all 20 participants.

The simulated Q-learners were recovered as plain Q-learners with $\kappa \approx 0$. The perseveration model therefore did not win merely because of its extra parameter.

3.2 Bandit: matched human design



Learning curves when the LLMs played the human task design (two games of 24 trials, 75% vs 25%).

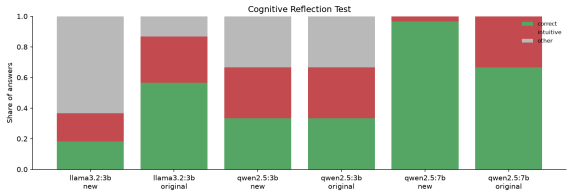
Table 2. Behaviour on the matched human design (medians per participant).

Group	Accuracy	Switch rate	Best model (count)	κ
Humans ($n = 44$)	0.82	0.21	Q 26 of 44	0.38
llama3.2:3b ($n = 20$)	0.50	0.35	random 14, Q+P 5, Q 1	0.62
qwen2.5:3b ($n = 20$)	0.58	0.09	Q 10, Q+P 10	1.61
qwen2.5:7b ($n = 20$)	0.90	0.11	Q+P 13, Q 7	1.69

llama3.2 still did not learn (accuracy vs humans: U = 803, $p < 10^{-6}$), and random was its best model for 14 of 20 participants. qwen2.5:3b behaved differently on this easier task. Its participants fell into three groups. Nine of 20 reached 75–100% correct, three locked onto the worse machine (about 2% correct), and eight stayed near chance. Because of this bimodality, its median accuracy did not differ significantly from the human median (U = 517, $p = .27$), and half its participants were best fitted by plain Q-learning. It remained far stickier than humans (switch rate: U = 683, $p < .001$; κ: U = 242.5, $p = .004$). Whatever qwen2.5:3b settled on early, right or wrong, it kept.

qwen2.5:7b reached a median of 90% correct, numerically above the human median (U = 557, $p = .09$). Fifteen of its 20 participants were above 80% correct, none locked onto the worse machine, and four stayed near chance. Its stickiness, however, was unchanged from the 3B model and remained above the human level (switch rate: U = 227.5, $p = .002$; κ: U = 235, $p = .003$).

3.3 Cognitive Reflection Test



CRT performance on original and new items.

Table 3. CRT results. Each item was presented 10 times.

Group	Originals correct (of 3)	New: correct	New: intuitive	New: other
Humans (Frederick, 2005)	1.24	—	—	—
llama3.2:3b	1.70	18%	18%	63%
qwen2.5:3b	1.00	33%	33%	33%
qwen2.5:7b	2.00	97%	3%	0%

llama3.2 exceeded the human average on the original items but solved only 18% of the new ones. Most of its errors on new items were not the intuitive answer. They were other numbers, often far off (for example, 791 cents for an item whose answer is 50). qwen2.5:3b's even split on the new items (33% correct, 33% intuitive, 33% other) is an artefact of determinism rather than a balanced mix of strategies. Although each of the 10 samples used a different seed at

temperature 0.7, qwen2.5 gave the same answer every time on 8 of 9 items. Its 60 answers to new items therefore reduce to 6 item-level outcomes: 2 correct, 2 intuitive and 2 other, which gives exactly one third each. The item-level answers show a further problem. qwen2.5 answered "50" on all three cost items (8 of 10 times for bat-and-ball, whose answer is 5; always for notebook-and-pen, whose answer is 10; and always for pizza-and-drink, whose answer happens to be 50). Its one "correct" cost item is therefore a coincidence of a fixed default answer, not reasoning. Excluding it, qwen2.5 solved 1 of 6 new items (cooks), and on the doubling items it gave the intuitive answer every time.

qwen2.5:7b solved almost all new items (97%; its only errors were 2 of 10 intuitive answers on pizza-and-drink) and two of the three originals. Strikingly, it answered the original bat-and-ball problem with the intuitive "10 cents" on all 10 samples, while correctly solving the two new items with the same structure.

4. Discussion

Reward learning appears with scale, but perseveration does not go away. Even with the complete history of outcomes in front of them, neither 3B model used reward to guide its choices in the way humans do. The two models failed in qualitatively different ways. llama3.2 ignored reward and chose close to randomly. qwen2.5:3b committed to an option and kept it. That looks more like a strong habit than like exploitation of a learned value, because a learner that exploits still abandons an option that keeps losing. The matched-design runs show that these failures are not explained by the harder LLM task. When the better option was easy to find, qwen2.5:3b's habit sometimes locked it onto the right machine and sometimes onto the wrong one. This produced a group median close to the human median but a very different pattern for individual participants.

Moving from 3B to 7B within the same model family changed this picture. qwen2.5:7b learned which option was better, and on the human design its accuracy was at least as high as the humans'. Its stickiness, however, did not change: κ was about 2 in both sizes, roughly five times the human value. The larger model therefore did not replace the habit with reward learning but added reward learning on top of it. A high κ combined with accurate choices means that it exploits well but rarely explores, which would be costly in a changing environment.

These results contrast with the good bandit performance of GPT-3 reported by Binz and Schulz (2023), and with the structured in-context learning of larger models (Schubert et al., 2024). Together with the 7B result, they suggest that learning from reward in context emerges somewhere between 3B and 7B parameters in this model family. The two 3B models failed in different ways, so training details also matter at this scale.

Perseveration must be modelled. Without the perseveration model, the first analysis labelled qwen2.5 as "random", which hides the most distinctive structure in its behaviour. Perseveration is a standard component of human and animal choice models (Lau & Glimcher, 2005; Katahira, 2015). These results suggest that it should be a default part of any model set applied to LLMs, because a stickiness term can absorb behaviour that would otherwise be attributed to learning or to noise.

CRT performance reflects familiarity rather than reflection. llama3.2's drop from above-human performance on the famous items to 18% on new versions with the same structure fits memorisation of training data. This matches broader evidence that LLM performance depends on how familiar a problem's surface form is (McCoy et al., 2023). Its errors on the new items were mostly neither reflective nor intuitive, so it is not "intuitive like people". It is wrong in ways that people seldom are. qwen2.5:7b shows the opposite dissociation. It reasons correctly about new versions of the problems, yet on the most famous item it reproduces the famous *wrong* answer. One possible explanation is that the bat-and-ball problem appears in training text mostly as an example of the intuitive error, so the model has learned the surface pattern "bat and ball → 10 cents" more strongly than the reasoning. Either way, performance on original CRT items is not a reliable measure of LLM reasoning in either direction, and new items with the same structure are needed.

qwen2.5:3b's answers were almost fully deterministic across seeds, and its apparent success on one item came from a fixed default answer that happened to be correct. Studies that treat repeated samples from one model as independent participants should therefore check whether the samples really differ, report results per item, and include items whose correct answers differ from the model's preferred default.

5. Limitations

First, only three models were tested, and the comparison of scale rests on a single pair (qwen2.5 at 3B and 7B), so the emergence of reward learning cannot yet be placed precisely or generalised to other model families. Second, the human and LLM tasks differ in one remaining respect: humans visited their casinos interleaved, while the LLMs played each game in one block. Third, the LLMs saw the full history as text while humans had to remember it. This should favour the

models, yet they still performed worse. A prompt that summarises each option's payouts could separate a failure to reason from reward from a failure to read a long history. Fourth, human α and β were poorly recovered with 48 trials, so humans were compared with the models mainly on model-free measures, best-model counts and κ, which recovered well. Fifth, "participants" from one LLM are samples from a single model, not independent individuals, so the group statistics describe variability in sampling, not between people. Finally, the CRT human reference is a published mean rather than trial-level data, and the six new items have not been normed on human participants.

Future work should test more model sizes and families to locate where learning from reward emerges and whether perseveration ever declines, interleave the LLM games to match the human design fully, and apply the same pipeline to the other bandit experiments in Psych-101.

Data and code availability

All code, prompts, raw choices, fitted parameters and figures are available at https://github.com/Muzz698/llm-cognitive-subjects. The repository includes automated tests, and every result can be reproduced on a laptop with free software.

Author contributions and use of AI tools

Muzamil conceived the study, chose the models and tasks, ran all experiments on local hardware, decided which analyses to add (the perseveration model, parameter recovery and the matched human design), checked every result against the raw data, and wrote and approved the final manuscript. An AI assistant (Claude, Anthropic) was used as a programming and writing tool, to implement analysis code to the author's specification and to edit the text. All scientific decisions and interpretations are the author's, and the author takes full responsibility for the content.

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